CENTRAL UNIVERSITY

DESIGN AND IMPLEMENTATION OF AN INTEGRATED DIGITAL FINANCIAL MANAGEMENT AND BUSINESS INTELLIGENCE SYSTEM FOR RETAIL AND TRADING ENTERPRISES IN SIERRA LEONE: A CASE STUDY OF A SELECTED COMPANY IN FREETOWN

FINBIZ 360

A project proposal submitted in partial fulfilment of the requirements for the award of the Bachelor of Science (Honours) in Business and Information Technology

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Industrial Organisation: ……..
Location: Freetown, Sierra Leone
Academic Year: 2025/2026

DRAFT FOR SUPERVISOR AND ORGANISATION REVIEW

DOCUMENT CONTROL

Field	Description
Document	FinBiz 360 – Industrial Project Proposal and System Requirements Specification
Programme	BSc Honours Business and Information Technology
Version	Draft 1.0
Study site	One selected retail or trading company in Freetown
Payment channels	Cash, mobile money and bank transfers
Research access	Authorised transaction records, staff interviews and observation
Duration	20 weeks, proposed
Approval status	Not yet approved; subject to university and organisation review

ABSTRACT

Retail and trading enterprises may receive payments through cash, mobile money and bank transfers. When records are maintained in separate notebooks, spreadsheets or provider statements, management may face delayed reconciliation, inconsistent financial information, limited cash-flow visibility and difficulty producing timely reports. This project proposes FinBiz 360, an integrated digital financial management and business intelligence system for one selected retail or trading company in Freetown, Sierra Leone.

The proposed system will provide centralised transaction recording, customer and supplier account management, income and expense monitoring, statement import, transaction matching and reconciliation, approval controls, audit trails and management dashboards. Advanced capabilities will include rule-based anomaly indicators and short-term cash-flow forecasting, subject to data sufficiency. The study will use a single-organisation case study, mixed-methods data collection and Design Science Research Methodology. Data will be gathered through authorised document review, staff interviews, operational observation and user evaluation. The system will be assessed using transaction accuracy, reconciliation time, match rates, usability, response time and forecast error.

The expected outputs are a working enterprise application, database, technical documentation, user guide, evaluation report and a research manuscript prepared for possible publication. FinBiz 360 is scoped as an internal financial management and reconciliation platform. It will not hold customer funds, independently settle payments or claim to be a licensed payment service.

Keywords: FinTech; digital financial management; business intelligence; transaction reconciliation; SMEs; cash flow; Sierra Leone.

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

Digital financial services have changed how businesses receive payments, transfer money and maintain financial records. Retail and trading enterprises may receive cash at the point of sale, mobile money transfers from customers and bank transfers from business clients or suppliers. The use of multiple channels creates convenience but also increases the need to maintain consistent transaction records and reconcile internal entries against independent payment evidence.

Where financial processes are fragmented across spreadsheets, notebooks and provider statements, staff may compare records manually. This can delay daily summaries, make missing or duplicated entries difficult to identify and limit management's ability to obtain a consolidated view of income, expenses and available cash. These issues connect directly to Business and Information Technology through business process design, information quality, database management, internal controls and decision support.

This project will investigate the existing financial management practices of one retail or trading company in Freetown. Findings will inform the design and implementation of FinBiz 360. The platform will centralise transaction records, support statement-based reconciliation for mobile money and bank transfers, provide cash transaction controls, monitor income and expenses, and present management information through dashboards and reports.

The system is intended for internal organisational use. It will record transactions and evidence supplied by authorised users or approved interfaces. It will not independently provide payment services, custody funds or initiate settlement. Any future provider API connection will require official documentation, sandbox or production access, written authorisation and review of applicable obligations.

1.2 Statement of the Problem

The selected company may use different processes to record cash receipts, mobile money payments and bank transfers. Where these records are not consolidated, staff may need to compare internal transaction logs with external statements manually. The organisation may consequently experience difficulty identifying unmatched transactions, duplicate references, differences in recorded amounts, outstanding customer balances and delayed expense entries.

Fragmented financial information can reduce the timeliness and reliability of management reports. It may also make it harder to monitor cash flow, compare actual performance against budgets and identify unusual transaction patterns. The extent and causes of these issues must not be assumed; they will be established through interviews, observation and authorised review of records.

The problem addressed is the absence or limited availability of a single, controlled information system that brings together multi-channel transaction records, reconciliation, financial monitoring and management reporting for the selected organisation. FinBiz 360 will be designed and evaluated as a context-specific response to verified requirements.

1.3 Aim of the Study

To design, implement and evaluate an integrated digital financial management and business intelligence system that improves transaction management, financial reconciliation, cash-flow monitoring and management decision support in a selected retail or trading company in Freetown, Sierra Leone.

1.4 Specific Objectives

1. Examine the existing financial transaction and record-management processes of the selected company.
2. Identify operational, information-quality, reporting and control challenges associated with cash, mobile money and bank transfers.

3. Analyse the business, functional, security and technical requirements of an integrated digital financial management system.
4. Design a secure enterprise architecture and relational database for centralised financial transaction management.
5. Develop modules for transaction recording, customer and supplier accounts, income and expenses, statement import, reconciliation and management reporting.
6. Implement audit trails, role-based access, approval controls and rule-based indicators for potentially unusual transactions.
7. Evaluate the system using transaction accuracy, reconciliation efficiency, usability, performance and cash-flow forecasting error.

1.5 Research Questions

8. What processes does the selected organisation currently use to record and monitor cash, mobile money and bank transfer transactions?
9. What operational and information-management challenges arise when records from different payment channels are maintained separately?
10. How can a centralised system support the recording, validation and reconciliation of multi-channel transactions?
11. To what extent does the proposed system affect the time and accuracy of the organisation's reconciliation process?
12. How useful are the system dashboards and reports to staff and management in monitoring financial performance and cash flow?
13. How do intended users assess the system's usability, reliability and suitability for the organisation's workflow?

1.6 Proposed Hypotheses (if required)

- H1: FinBiz 360 reduces the time required to reconcile eligible transaction records compared with the organisation's baseline process.
- H2: FinBiz 360 improves the proportion of eligible transactions recorded with complete and internally consistent information.
- H3: Intended users report positive perceived usefulness and satisfaction after completing tasks with FinBiz 360.

These are hypotheses to be tested, not claims of achieved results. The final statistical approach will depend on sample size, measurement design and supervisor guidance.

1.7 Significance of the Study

- Participating organisation: may gain consolidated records, structured reconciliation, timely reporting and clearer accountability.
- Management: may gain improved visibility into income, expenses, outstanding balances and cash-flow trends.
- Employees: may benefit from clearer transaction-entry procedures and reduced repetitive manual comparison.
- BIT discipline: demonstrates the application of systems analysis, software engineering, database management and business intelligence to an industrial problem.
- Academic community: provides a documented case study and evaluation relevant to digital financial management in SMEs.
- Student researcher: gains practical experience in requirements elicitation, enterprise development, testing, deployment and research communication.

1.8 Scope of the Study

1.8.1 Organisational scope

The study will involve one retail or trading company in Freetown. The organisation's name, business profile, participating departments and permission arrangements will be inserted after formal confirmation.

1.8.2 Functional scope

- User authentication, roles and permissions.
- Customer, supplier and account records.
- Recording cash, mobile money and bank transfer transactions.
- Income, expenses, invoices and payment status management.
- Import of authorised mobile money and bank statements using supported file formats.
- Rule-based matching, duplicate checks, exception queues and reconciliation reports.
- Management dashboards, financial summaries and controlled exports.
- Audit logs, approval workflows and backup procedures.
- Exploratory cash-flow forecasting and anomaly indicators, subject to data sufficiency.

1.8.3 Exclusions

- No customer funds custody, wallet operation, clearing or independent payment settlement.
- No storage of mobile money PINs, bank passwords, card CVVs or authentication secrets.
- Live payment-provider API integration is conditional on access, technical documentation and written authorisation.
- The project does not replace a regulated bank, payment service provider or the organisation's statutory accounting obligations.
- Tax filing and full ERP functions are outside the core scope unless formally approved as additions.

1.9 Limitations and Delimitations

The study concerns one organisation, so findings may not generalise to all SMEs in Sierra Leone. Historical records may be incomplete or inconsistent. Provider APIs may be inaccessible. Forecasting and anomaly detection depend on the volume and quality of available data. These limitations will be documented, and masked or synthetic test data may be used where live data access is restricted.

1.10 Operational Definitions

Term	Meaning in this project
FinTech	Use of digital technology to support or improve financial services and processes.
Digital financial management	Electronic recording, control, analysis and reporting of organisational financial activities.
Transaction reconciliation	Comparing internal transaction records with independent statements or evidence to identify matches and exceptions.
Payment channel	Method through which a payment is made or received, such as cash, mobile money or bank transfer.
Business intelligence	Tools and processes that transform business data into reports, dashboards and analytical insight.
Anomaly indicator	A rule- or model-generated signal requiring human review; not proof of fraud.
Cash-flow forecast	Estimate of future cash inflows and outflows based on historical data and stated assumptions.

1.11 Organisation of the Report

The final report will contain five chapters: Chapter One introduces the study; Chapter Two reviews literature, concepts and theoretical foundations; Chapter Three explains the research and development methodology; Chapter Four presents system design, implementation and testing; and Chapter Five discusses findings, conclusions, recommendations and future work.

CHAPTER TWO: LITERATURE REVIEW – PROPOSED FRAMEWORK

The final literature review should critically examine verified scholarly and authoritative sources, not merely describe technologies.

2.1 Digital Financial Management in SMEs

Review SME use of digital tools for transaction recording, cash management, receivables, supplier payments and reporting. Examine barriers such as cost, skills, connectivity, trust and fragmented data.

2.2 Financial Information Systems and Internal Controls

Examine transaction processing systems, accounting information systems, segregation of duties, approval controls, audit trails, data integrity and reliable source records.

2.3 Multi-Channel Payment Management

Discuss cash, mobile money and bank transfer records, including identifiers, settlement timing, reversals, fees, missing references and provider-specific statement formats.

2.4 Transaction Reconciliation

Review exact-reference matching, amount/date matching, tolerance rules, one-to-many and many-to-one matching, duplicate detection, exception management and human review. A suggested match must remain distinguishable from a confirmed reconciliation.

2.5 Business Intelligence and Decision Support

Review dashboards, KPIs, reporting latency, drill-down analysis, budget-versus-actual reporting and the role of information quality in managerial decisions.

2.6 Cash-Flow Forecasting and Anomaly Detection

Review baseline forecasting methods, moving averages and appropriate time-series approaches. Discuss rule-based alerts and statistical anomaly detection, including false positives, data requirements, explainability and human review.

2.7 Theoretical Foundations

Potential frameworks include the DeLone and McLean Information Systems Success Model for system quality, information quality and user satisfaction; the Technology-Organisation-Environment framework for adoption context; and internal-control principles for accountability. Select and justify the framework that best aligns with the research questions.

2.8 Research Gap

The study will investigate the practical gap between the use of multiple payment channels and the availability of a unified, auditable management view in one SME. The specific gap must be established through a critical literature review and evidence about the organisation's existing process.

CHAPTER THREE: RESEARCH AND DEVELOPMENT METHODOLOGY

3.1 Research Design

The project will use an applied single-organisation case study with mixed methods. Qualitative evidence will explain existing workflows, user needs and constraints. Quantitative evidence will support baseline measurement and system evaluation. Design Science Research Methodology will guide construction and evaluation of the FinBiz 360 artefact.

3.2 Design Science Research Activities

Activity	Application	Evidence / output
Problem identification	Study current transaction recording, reconciliation and reporting.	Process maps, interviews, baseline findings.
Objectives definition	Translate verified problems into measurable goals.	Objectives and acceptance criteria.
Design and development	Create architecture, database, interfaces and modules.	Design artefacts, source code and prototype.
Demonstration	Run business scenarios using authorised data.	Demonstration scripts and results.
Evaluation	Test correctness, usability, performance and reconciliation.	Test report, metrics and feedback.
Communication	Present findings and technical contribution.	Dissertation, presentation and manuscript.

3.3 Study Site and Population

The site will be one retail or trading company in Freetown. The population may include the owner/manager, accountant or finance officer, cashier or sales personnel, store or operations staff and other employees who create, review or use financial records. Final population and sample will be determined after access permission and a staff list are obtained.

3.4 Sampling Strategy

Purposive sampling is appropriate for participants with direct knowledge of financial processes. In a small organisation, all relevant staff may be invited; in a larger organisation, participants may be selected by role. An initial planning range of 10–20 participants may be considered, but the final sample must be justified by workforce size, research questions, access and supervisor guidance.

Participant group	Indicative range	Information sought
Owner / manager	1–2	Reporting, controls, decisions and approval rules.
Accountant / finance officer	1–3	Cashbook, bank/mobile money records, expenses and reconciliation.
Cashier / sales staff	2–6	Sales recording, receipts and payment-channel workflow.
Store / operations staff	1–4	Operational links to sales, stock and supplier payments.
Other relevant users	As applicable	Reporting, review and usability needs.

3.5 Data Collection Methods

- Semi-structured interviews with the owner, manager and relevant staff.
- Direct observation of authorised transaction recording and reconciliation.
- Document review of approved, redacted sales logs, cashbooks, invoices, expenses and payment statements.
- Questionnaires or task-based usability instruments for intended users.
- System test logs and benchmark measurements during evaluation.

3.6 Data Analysis

Qualitative evidence will be coded into themes such as transaction capture, reconciliation, reporting, approval, exception handling and usability. Quantitative data will be summarised using counts, percentages, averages and suitable dispersion measures. For paired baseline and post-implementation observations, an appropriate paired test may be considered after checking assumptions and sample size. Effect sizes and limitations should be reported.

3.7 Development Method

An iterative Agile-inspired process will be used. Core transaction and access-control features will be built first, followed by reconciliation, reporting and advanced analytics. Each iteration will be demonstrated to relevant users where practical, with feedback recorded in a prioritised backlog.

3.8 Ethical Considerations and Data Protection

- Obtain written company permission and required university ethics approval before data collection.
- Explain the study and obtain informed consent from interview and questionnaire participants.
- Use only authorised records; redact customer names, account details and personal identifiers where possible.
- Do not collect PINs, passwords, CVVs or authentication secrets.
- Use synthetic or masked records for demonstrations unless live data use is explicitly authorised.
- Store research files securely, restrict access and follow the approved retention and deletion plan.
- Do not publish commercially sensitive information or identify the company without written consent.
- Document the researcher's role and any potential conflict of interest.

3.9 Validity, Reliability and Trustworthiness

Triangulate interviews, observation and document review. Validate requirements with users. Use repeatable test cases linked to requirements. Review and pilot questionnaires where appropriate. Document data cleaning and exclusion rules.

3.10 Evaluation Metrics

Metric	Definition / calculation	Evidence
Reconciliation rate	Matched eligible statement lines ÷ total eligible statement lines × 100.	Reconciliation batch results.
Reconciliation time	Elapsed time from batch start to completion of review.	Baseline and system task timings.
Record completeness	Records with all required fields ÷ records tested × 100.	Data-quality test.
Duplicate precision	True duplicate alerts ÷ all duplicate alerts × 100.	Labelled test data.
Duplicate recall	True duplicate alerts ÷ all known duplicates × 100.	Labelled test data.
Response time	Time to complete defined requests under stated conditions.	Performance test logs.
Usability	Task completion, error counts and user questionnaire scores.	User evaluation.
Forecast MAE	Mean absolute error between forecast and actual cash-flow values.	Chronological holdout dataset.

Define metrics before testing. Distinguish automatically matched items from those resolved after human review. For anomaly detection, report false positives and false negatives where labelled test data exist.

DETAILED SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SRS 1. Purpose and Product Perspective

FinBiz 360 is a browser-based internal enterprise information system for one retail or trading organisation. It centralises transaction records and supports financial monitoring and reconciliation. It interacts with users through a web interface and ingests authorised statement files. Provider integration is optional and conditional on access approval.

SRS 2. User Classes and Roles

Role	Core permissions

System Administrator	Manage users, roles, system configuration, reference data and backups; no automatic business approval authority unless separately assigned.
Owner / Manager	View management dashboards and reports; approve designated transactions, adjustments and exceptions.
Finance Officer	Record and review transactions, import statements, reconcile records and prepare reports.
Cashier / Sales Officer	Record sales and receipts within assigned permissions; view permitted transactions.
Auditor / Read-only Reviewer	View authorised records, reconciliation evidence and audit history without changing records.

SRS 3. Functional Requirements

ID	Requirement	Description	Priority
FR-01	Authentication	Authenticate registered users before protected access.	Must
FR-02	Role-based access	Enforce permissions according to role and action.	Must
FR-03	User administration	Create, update, deactivate and assign user roles.	Must
FR-04	Customer management	Maintain customer records with minimum necessary data.	Must
FR-05	Supplier management	Maintain supplier records.	Must
FR-06	Account setup	Configure cash, bank, mobile money clearing and income/expense categories.	Must
FR-07	Transaction capture	Record date, amount, currency, direction, channel, reference, description and responsible user.	Must
FR-08	Cash transactions	Record cash receipts and payments with receipt/voucher references.	Must
FR-09	Mobile money records	Record mobile money details and external references from authorised evidence.	Must
FR-10	Bank transfer records	Record bank transfer details and external references from authorised evidence.	Must
FR-11	Sales and invoices	Record sales/invoices, customer, amount, due date and payment status.	Should
FR-12	Expense recording	Record category, payee, amount, date, evidence and approval status.	Must
FR-13	Statement import	Import supported CSV/XLSX statements using preview and validation.	Must
FR-14	Import validation	Identify invalid rows, missing fields, unsupported formats and possible duplicate imports.	Must
FR-15	Transaction matching	Suggest matches using reference, amount, date, channel and configured rules.	Must
FR-16	Exception queue	List unmatched, partially matched, duplicate and discrepant items.	Must
FR-17	Manual resolution	Allow authorised users to document and approve manual match, split, fee or exception decisions.	Must
FR-18	Reconciliation report	Report matched, unmatched, pending and exception items by channel and period.	Must
FR-19	Cash-flow view	Summarise recorded inflows, outflows and balances by account and date range.	Must
FR-20	Dashboard	Present authorised KPIs for volume, income, expenses, outstanding items and reconciliation status.	Must

FR-21	Audit log	Record significant actions, timestamps, affected records and relevant before/after values.	Must
FR-22	Approval workflow	Route designated transactions, adjustments and write-offs for approval by configured rules.	Should
FR-23	Exports	Export permitted reports to PDF or spreadsheet formats.	Should
FR-24	Notifications	Notify authorised users about unresolved exceptions and pending approvals.	Could
FR-25	Anomaly indicators	Flag unusual patterns for human review and record the triggering rule.	Should
FR-26	Cash-flow forecasting	Generate short-term forecasts where sufficient reliable data exist and display assumptions.	Could
FR-27	Backup and restore	Perform and verify scheduled backups and controlled restoration.	Must
FR-28	Period locking	Allow authorised reporting-period closure and controlled reopening with reason.	Should

SRS 4. Business Rules

- Store monetary values using fixed-precision decimal types and explicit currency codes; confirm the organisation's currency configuration during discovery.
- Each transaction has a unique internal identifier. External references are separate and are not assumed globally unique across providers.
- Statement imports retain source-file metadata, import time, uploader and validation result.
- A transaction is not reconciled solely because its amount is similar; the rule and evidence must be recorded.
- Automatic matches are explainable and reversible through an authorised, audited exception workflow.
- Completed financial records are not silently deleted or overwritten; corrections use reversals, adjustments or amendments with reasons and approvals.
- Recorded, provider-confirmed and statement-reconciled are distinct transaction states.
- Fees, reversals, refunds, partial payments and settlement differences are represented explicitly where supported by source records.
- Only authorised roles approve adjustments, write-offs and period closure.
- Payment credentials such as PINs, passwords and CVVs are never stored.

SRS 5. Non-Functional Requirements

ID	Requirement
NFR-01 Security	HTTPS in deployed environments, secure password hashing and server-side role-based access.
NFR-02 Data integrity	Database constraints, validation and atomic transactions protect financial consistency.
NFR-03 Auditability	Critical actions are attributable to a user and timestamp; ordinary users cannot alter audit records.
NFR-04 Performance	Common queries should meet an agreed target; provisional target is 3 seconds at the 95th percentile for the defined test environment.
NFR-05 Usability	Core tasks should be usable after brief training; task completion and error rates will be measured.
NFR-06 Availability	Documented backup, restore and outage procedures appropriate to the organisation.
NFR-07 Scalability	Support growth through indexed queries and modular services.
NFR-08 Maintainability	Version-controlled, documented source code separated into logical modules.
NFR-09 Compatibility	Support current major desktop browsers and responsive common mobile screens.
NFR-10 Privacy	Collect only necessary data; mask or anonymise research exports where feasible.

NFR-11 Recoverability	Agree and test backup frequency, recovery point objective and recovery time objective.
NFR-12 Explainability	Anomaly alerts and reconciliation suggestions show the rule or reason behind the result.

SRS 6. External Interface Requirements

- User interface: responsive browser application with dashboards, searchable tables, filters, forms and validation messages.
- Statement interface: controlled CSV/XLSX import with configurable mapping, preview, validation, duplicate-file detection and error report.
- Optional provider API: server-to-server integration only with official documentation, sandbox or production credentials and written permission.
- Reporting interface: controlled exports with date, channel, status and user filters.
- Database interface: controlled application data access; reporting uses restricted views or a separate analytics dataset.

SRS 7. Acceptance Criteria

Area	Minimum acceptance evidence
Authentication	Unauthorised users cannot access protected modules; role tests pass.
Transaction capture	Valid cash, mobile money and bank records save correctly; invalid records are rejected.
Statement import	Valid sample files import; invalid rows and duplicate imports are handled.
Reconciliation	Known matched, unmatched, duplicate and amount-difference cases produce expected outcomes.
Audit trail	Critical create, approve, reverse and reconcile actions are traceable.
Dashboard	Displayed totals agree with underlying test records for the same filters.
Backup	A test backup can be restored and verified.
User evaluation	Intended users complete agreed tasks and provide documented feedback.

SRS 8. Release Priorities

Release 1: secure access, master data, transaction capture, statement import, basic matching, exception handling, audit logging and core reports. Release 2: approval workflows, advanced dashboards and cash-flow analysis. Release 3: anomaly indicators and forecasting only if data quality and time permit.

UML AND SYSTEM DESIGN

The following editable Mermaid source represents the proposed logical design. Update it after requirements validation. Mermaid source may be rendered using a compatible Mermaid editor for the final dissertation.

UML 1. Use Case View

```
flowchart LR

  Admin([System Administrator])

  Manager([Owner / Manager])

  Finance([Finance Officer])

  Cashier([Cashier / Sales Officer])

  Auditor([Auditor / Read-only Reviewer])
```

```
System((FinBiz 360))

Admin -->|Manage users and settings| System

Manager -->|View dashboards and reports| System

Manager -->|Approve adjustments and exceptions| System

Finance -->|Record transactions and expenses| System

Finance -->|Import statements| System

Finance -->|Reconcile transactions| System

Cashier -->|Record sales and receipts| System

Auditor -->|Review records and audit trail| System
```

Interpretation: the administrator manages access and configuration; the manager reviews performance and approves designated actions; finance staff maintain and reconcile records; cashiers capture sales and receipts; auditors have controlled read-only access.

UML 2. Core Domain / Class Diagram

```
classDiagram

class User { +UUID userId; +string fullName; +string email; +bool active }

class Role { +UUID roleId; +string roleName }

class Organisation { +UUID organisationId; +string name; +string location }

class Account { +UUID accountId; +string accountName; +string accountType; +string currencyCode }

class PaymentChannel { +UUID channelId; +string channelName; +string providerName }

class Transaction { +UUID transactionId; +string externalReference; +decimal amount; +date transactionDate; +string
direction; +string status }

class Statement { +UUID statementId; +string fileName; +datetime importedAt }

class StatementLine { +UUID lineId; +string externalReference; +decimal amount; +date valueDate }

class Reconciliation { +UUID reconciliationId; +string matchStatus; +string matchRule; +datetime resolvedAt }

class AuditLog { +UUID auditId; +string action; +datetime createdAt }

class Approval { +UUID approvalId; +string decision; +string reason; +datetime decidedAt }

Role "1" --> "*" User : assigned

Organisation "1" --> "*" User : employs

Organisation "1" --> "*" Account : owns

Account "1" --> "*" Transaction : records

PaymentChannel "1" --> "*" Transaction : used by

Statement "1" --> "*" StatementLine : contains

Transaction "1" --> "*" Reconciliation : candidate

StatementLine "1" --> "*" Reconciliation : candidate

User "1" --> "*" AuditLog : performs

User "1" --> "*" Approval : decides
```

Final implementation should include explicit journal entries and journal lines if accounting-grade ledger functionality is required. A balanced double-entry model must enforce total debits equal total credits for each posted journal.

UML 3. Sequence – Statement Import and Reconciliation

```
sequenceDiagram
 actor FinanceOfficer
 participant UI as FinBiz Web App
 participant API as Django API
 participant Import as Import Validator
 participant DB as PostgreSQL
 participant Engine as Reconciliation Engine
 FinanceOfficer->>UI: Upload statement
 UI->>API: Submit file and channel
 API->>Import: Validate format and rows
 Import->>DB: Check import fingerprint and references
 DB-->>Import: Duplicate and validation results
 Import-->>API: Preview valid and invalid rows
 API-->>UI: Show preview and errors
 FinanceOfficer->>UI: Confirm import
 UI->>API: Commit approved import
 API->>DB: Save statement and lines
 API->>Engine: Generate match suggestions
 Engine->>DB: Compare internal transactions
 DB-->>Engine: Candidate records
 Engine->>DB: Save suggestions and exceptions
 Engine-->>API: Matching summary
 API-->>UI: Show matched and unmatched items
 FinanceOfficer->>UI: Review exception
 UI->>API: Submit authorised resolution
 API->>DB: Save resolution and audit event
 API-->>UI: Updated reconciliation status
```

UML 4. Activity – Transaction Lifecycle

```
flowchart TD
 A([Start]) --> B[Capture transaction]
 B --> C{Required fields valid?}
```

```
C -->|No| D[Show validation errors]

D --> B

C -->|Yes| E[Save as recorded / pending]

E --> F{Payment evidence available?}

F -->|No| G[Await evidence or follow up]

F -->|Yes| H[Record confirmation evidence]

G --> H

H --> I[Import or retrieve statement line]

I --> J[Run matching rules]

J --> K{Match and amount consistent?}

K -->|Yes| L[Mark reconciled]

K -->|No| M[Create exception]

M --> N[Authorised staff review]

N --> O{Resolution approved?}

O -->|No| M

O -->|Yes| P[Record resolution and audit trail]

P --> L

L --> Q[Update reports and dashboard]

Q --> R([End])
```

UML 5. Component Architecture

```
flowchart TB

 subgraph Client

  Web[React Web Interface]

 end

 subgraph Application

  API[Django REST API]

  Auth[Identity and Access Control]

  Tx[Transaction Service]

  Recon[Reconciliation Service]

  Finance[Financial Management Service]

  Audit[Audit and Approval Service]

  Reports[Reporting Service]

 end

 subgraph Data

  DB[(PostgreSQL)]
```

```
 Files[(Controlled Statement File Storage)]
end
subgraph Analytics
 ETL[Analytics Data Pipeline]
 ML[Forecasting and Anomaly Analysis]
 BI[Power BI / Dashboard]
end
Web --> API
API --> Auth
API --> Tx
API --> Recon
API --> Finance
API --> Audit
API --> Reports
Tx --> DB
Recon --> DB
Finance --> DB
Audit --> DB
Reports --> DB
Recon --> Files
DB --> ETL
ETL --> ML
ETL --> BI
```

UML 6. Deployment View

```
flowchart LR
 Browser[Staff Browser]
 HTTPS[HTTPS Reverse Proxy]
 App[Application Server]
 DB[(PostgreSQL Database)]
 Store[(Encrypted Backup / File Store)]
 Analytics[Analytics Environment]
 Browser --> HTTPS --> App
 App --> DB
 App --> Store
 DB --> Analytics
```

DEVELOPMENT ROADMAP AND IMPLEMENTATION PLAN

1. Proposed 20-Week Schedule

Period	Phase	Activities	Deliverable
Weeks 1–2	Initiation	Confirm organisation, title, access, ethics and project plan.	Approved proposal and access letter.
Weeks 3–4	Industrial discovery	Interview staff, observe workflows, review authorised records and establish baseline.	Process maps and discovery report.
Weeks 5–6	Requirements	Validate user stories, requirements, scope and risks.	Approved SRS and backlog.
Weeks 7–8	System design	Create ERD, UML, wireframes, security model and schema.	Design specification and prototype.
Weeks 9–12	Core development	Build authentication, roles, master data, transactions, expenses and audit.	Core application release.
Weeks 13–15	Reconciliation and analytics	Build import, validation, matching, exceptions, reports and dashboard.	Reconciliation/reporting release.
Weeks 16–17	Testing	Unit, integration, security, performance and UAT.	Test report and defect log.
Week 18	Deployment and evaluation	Deploy approved pilot, train users and collect evaluation data.	Deployment and evaluation records.
Weeks 19–20	Documentation	Complete dissertation, user guide, technical docs and presentation.	Final submission package.

2. Suggested Development Sprints

Sprint	Scope	Definition of done
0	Repository, environments, conventions and initial skeleton.	Development environment ready.
1	Authentication, roles, organisation settings and audit foundation.	Role tests pass and actions are logged.
2	Customers, suppliers, accounts, categories and transaction capture.	Cash, mobile money and bank test records save correctly.
3	Expenses, invoices, approvals and financial summaries.	Business workflows and permissions pass.
4	Statement import, mapping, validation and duplicate control.	Valid and invalid statement files handled correctly.
5	Matching engine, exception queue and reconciliation reports.	Known matching scenarios produce expected results.
6	Dashboard, KPI views, exports and cash-flow summaries.	Dashboard totals tie to underlying records.
7	Anomaly indicators and forecasting prototype.	Advanced functions tested on suitable data; limitations documented.
8	Hardening, UAT, deployment, training and documentation.	Acceptance criteria and handover checklist completed.

3. Baseline and Post-Implementation Evaluation

Before deployment, measure the existing process using a defined sample of transaction batches. Record item count, channels, time spent, discrepancies and unresolved items. After training and familiarisation, repeat comparable tasks using FinBiz 360. Use similar transaction volumes and state differences between batches.

Measure	Baseline	Post-implementation
Reconciliation time	Observe and time current process for a defined batch.	Time equivalent task in FinBiz 360.
Record completeness	Audit required fields in current records.	Audit required fields in system records.
Exception handling	Count and classify unresolved differences.	Count and classify system exceptions and resolutions.
User experience	Interview or survey users about existing workflow.	Use same or comparable questions after training.
Reporting effort	Record time and steps to prepare a standard report.	Record time and steps to generate the system report.

4. Project Management and Communication

- Hold weekly progress reviews with the academic supervisor.
- Arrange discovery and validation meetings with the organisation's nominated contact.
- Maintain a requirements traceability matrix linking requirements, design, implementation and tests.
- Use a version-controlled backlog to track tasks, defects and change requests.
- Obtain written approval before changing project scope or accessing additional records.
- Maintain a decision log for key business and technical choices.

5. Roles and Responsibilities

Stakeholder	Responsibility
Student researcher/developer	Research, requirements, design, implementation, testing, documentation and presentation.
Academic supervisor	Academic guidance and review of research design, scope and deliverables.
Company sponsor/owner	Approve access, nominate users, validate needs and review suitability.
Finance representative	Explain transaction processes, statement formats and reconciliation rules.
End users	Participate in requirements validation, demonstrations, training and evaluation.
ICT contact, if available	Advise on hosting, access, security and deployment.

PROJECT RISK REGISTER

Risk	Likelihood / impact	Mitigation
Organisation access delayed	Medium / High	Secure sponsor and written permission early.
Financial records incomplete	High / High	Assess quality early; define exclusions and use masked test data.
Provider API unavailable	High / Medium	Make statement import the required integration path; API optional.
Scope too large	Medium / High	Prioritise recording and reconciliation; advanced analytics conditional.
Confidential information exposed	Low–Medium / High	Minimise data, mask identifiers, encrypt storage and restrict access.
False reconciliation	Medium / High	Explainable rules, confidence thresholds, human approval and reversible decisions.
Forecast model weak	Medium / Medium	Compare with baseline, disclose uncertainty and avoid unsupported claims.
Schedule delay	Medium / High	Use weekly milestones, prototypes and prioritised backlog.
User resistance	Medium / Medium	Involve users, provide training and collect feedback.
Outage or data loss	Low–Medium / High	Test backups, restoration and incident procedures.

SECURITY AND FINANCIAL CONTROL DESIGN

1. Identity and Access Management

- Use secure password hashing and session management.
- Enforce permissions at the server/API level, not only by hiding interface controls.
- Require stronger authentication for privileged roles where feasible.
- Deactivate accounts promptly when staff leave or change roles.
- Apply least privilege and review permissions periodically.

2. Financial Data Integrity

- Use database transactions for multi-step financial operations.
- Use fixed-precision decimal fields and explicit currency codes.
- Apply unique constraints to internal IDs and carefully scoped external references.
- Retain statement-file metadata and import evidence according to the approved retention policy.
- Use reversals and adjustments instead of silent edits to posted financial records.
- Maintain audit events for approvals, reversals, reconciliation decisions and period closures.

3. Reconciliation Controls

- Exact reference and amount matches may be proposed automatically where identifiers are reliable.
- Date/amount tolerance matches should be suggestions, not silently confirmed.
- Evaluate duplicate references within provider, channel, account and relevant date context.
- Represent fees, partial payments, grouped settlements and reversals explicitly.
- Require a reason for manual matching and independent approval for material items.
- Make reconciliation results reproducible from retained evidence and recorded rules.

4. Backup and Incident Handling

- Schedule encrypted backups and periodically test restoration.
- Monitor application errors, failed logins, unusual access and import failures.
- Maintain an incident register and escalation contact.
- Do not expose secrets or detailed internal errors to end users.
- Agree recovery objectives with the organisation before production deployment.